

# Spatial Audio Feedback for Hand Gestures in Games

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## ABSTRACT

We use game prototypes to investigate different audio feedback designs for hand gestures for encouraging immersion. Replacing explicit audio instructions for hand positions and movements with responsive audio feedback, suggestive of interaction methods by using environmental story-related audio cues, leads to measurably higher immersion. We present the design of responsive audio feedback for hand gestures in the context of such a game. The spatial organization of diegetic environmental feedback before, during, and after gesture input is supported by a structured approach towards the design of audio cues.

## CCS Concepts

•**Human-centered computing** → *Pointing devices; Auditory feedback; Sound-based input / output;*

## Keywords

immersion; audio-only; gesture interaction; touchless motion control; responsive feedback.

## 1. GESTURES IN AUDIO-ONLY GAMES

The availability of low-cost gesture detection hardware such as the Leap Motion allows game designers to apply and study the potential of touch-less motion control for more immersive spatial experiences. Norman [1] uses the term gestural interaction to subsume all different forms of motion-based interaction. O'Hara et al. [2] speak about touchless interaction. Here, the context of gestures, their embedding into the narrative of a game, and responsive feedback that reflects the space are particularly important. Audio-only games provide a unique opportunity to study what helps players in forming an inner picture of the environment.

Our approach is to experimentally compare designs of audio feedback that are responsive to hand gestures regarding immersion in gesture-controlled audio-only environments. For an experimental study, two design approaches for responsive

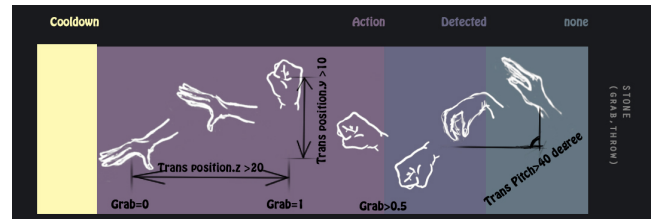


Figure 1: Illustration of a stone-throw gesture.

audio feedback were implemented, combining vocal instructions and non-diegetic audio cues on the one hand and environmental audio cues and diegetic (i.e. story-related) ones on the other.

We consider audio feedback not just as a result of gesture interaction, but as an interactive signal that guides the player while going through the whole process: before, during, and after the gesture sequence. The anticipation principle guides pre-gesture cues, setting up the expectation of an interaction with an object at a certain position, e.g. fish moving in a specific direction. Relative volume changes can further inform the player of the availability of a forthcoming gesture. During a gesture, the attachment of sound sources, such as noisy clothing or a bracelet with bells, can provide immediate feedback of the gesture in-progress. A real-time mapping of hand movement or gesture phases to audio parameters provides an opportunity for enhanced responsiveness that goes beyond replicating natural sounds. Fig. 1 shows the intermediate phases during one of the gestures in our testbed: grabbing and throwing a stone. After a gesture, noticeable positive or negative feedback, e.g. collisions with the virtual environment or more abstract feedback such as a "magic" watch changing its speed, guides further interaction.

These design principles have been successfully evaluated in an experimental setting with 26 subjects, showing a significant improvement in story immersion for the game version that uses the diegetic design.

## 2. REFERENCES

- [1] D. Norman. Natural user interfaces are not natural. *Interactions*, 17:6–10, 2010.
- [2] K. O'Hara, R. Harper, H. Mentis, A. Sellen, and A. Taylor. On the naturalness of touchless: Putting the 'interaction' back into NUI. *ACM Trans. Comput.-Hum. Interact.*, 20.

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